

 ML HACKATHON CHALLENGE

Industrial Chemical Reactor Yield Prediction

Developing a physics-informed machine learning surrogate model to predict Product B yield from operating conditions for real-time process optimization.

 Surrogate Model Architecture  Predictive Optimization  Machine Learning Pipeline



Project Overview & Context

Replacing computationally expensive physics-based numerical simulations with high-speed machine learning models to enable real-time chemical plant decision making.

Problem Context & Reaction Scheme

Simulation Challenge

Modern chemical manufacturing relies on intensive computational fluid dynamics (CFD) and heat transfer simulations. While accurate, these physics-based models are far too slow for real-time control and online optimization.

Solution: A data-driven machine learning surrogate model provides instant, accurate yield estimates directly from operational sensors.

Reaction Kinetics

The reactor governed by the competitive series reaction network:

Desired Reaction: $A \rightarrow B$

Side Reaction: $B \rightarrow C$

The ML model predicts the final **Overall Yield of Product B**, capturing complex nonlinear trade-offs between kinetics and heat transfer.

Methodology & End-to-End Workflow

-  **Exploratory Data Analysis & Quality Audit:** Evaluated structure, distributions, missing values, duplicates, and feature correlations across reactor operating parameters.
-  **Physics-Informed Feature Engineering:** Constructed domain features reflecting residence time, thermal gradients, and mass-flow rates to capture process kinetics.
-  **Multi-Model Regression Benchmarking:** Trained 5 gradient boosted and tree-based ensemble models (Random Forest, GBR, XGBoost, CatBoost, LightGBM).
-  **Explainability & Predictions:** Applied SHAP feature importance analysis to confirm chemical engineering validity and exported predictions on unseen test conditions.

Physics Feature Engineering



Temperature Difference

Captures jacket-to-inlet thermal driving force across the reactor vessel.

$$\Delta T = T_{\text{jacket}} - T_{\text{inlet}}$$



Residence Time Index

Proxy for reactant contact time inside the physical reactor vessel.

$$RTI = \frac{L_{\text{reactor}}}{Q_{\text{flow}}}$$



Thermal Energy Index

Interactions between inlet kinetic energy and cooling jacket capacity.

$$TEI = T_{\text{inlet}} \times T_{\text{jacket}}$$

Model Evaluation & Benchmark Results

Algorithm Model	Validation RMSE	R ² Score	Performance Rank
Gradient Boosting Regressor	13.6872	0.8935	🏆 Rank 1 (Best)
XGBoost Regressor	14.0719	0.8874	Rank 2
CatBoost Regressor	16.1259	0.8521	Rank 3
Random Forest Regressor	16.8892	0.8378	Rank 4
LightGBM Regressor	18.9658	0.7954	Rank 5

Predictive Model Accuracy (R²)



Key Finding: Gradient Boosting Regressor achieved the highest accuracy ($R^2 = 0.8935$, $\text{RMSE} = 13.6872$), outperforming all other models and capturing complex non-linear thermal dynamics.

Model Feature Interpretability

🕒 SHAP Insights

Feature importance analysis using SHAP values confirms chemical engineering domain alignment:

- ✓ **Thermal Energy Index:** Most critical factor driving reactor yield output.
- ✓ **Inlet Temperature:** Strong impact on reaction rate constants.
- ✓ **Residence Time Index:** Key control variable governing reactant conversion.



Key Impact & Performance Summary

89.3%

Variance Explained

High predictive power ($R^2 = 0.8935$) across validation scenarios.

13.68

Lowest RMSE

Achieved by Gradient Boosting surrogate model.

< 10ms

Inference Time

Enables real-time closed-loop chemical plant control.